SABLE HARBOR

Runtime estate calculated design register

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State: GENERATED SYNTHETIC DESIGN; no operating acceptance

Owner: Enterprise Technology Services / Finance

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Authority: September 11 runtime decisions; generated output does not create canon

Structured companion: enterprise/services/source/runtime_sites_2026-09-11.json; enterprise/services/source/runtime_capital_plan_2026-09-11.json

Source content SHA-256: `6f359aa710dd0ece6c253cea1c3312f81831b0be63782a6315269478e69c4771`. Regenerate with `python -m enterprise.runtime.report`. Do not edit calculated values in this derivative.

Current design boundary

Selected providers remain uncontracted, unreserved and uninstalled. The synthetic owned parcel is acquired/preconstruction, with 0% vertical construction and 0 kW commissioned. September 2026 actual service-operation history is not established.

Site	Planned facility / geography	State
Reno primary colocation	Tahoe-Reno, Nevada	PROVIDER_SELECTED_PROCUREMENT_PENDING
Boise independent recovery colocation	Boise, Idaho	PROVIDER_SELECTED_PROCUREMENT_PENDING
SABLE HARBOR Northern Nevada Data Center	Tahoe-Reno Industrial Center / USA Parkway industrial district, Storey County, Nevada	LAND_ACQUIRED_PRECONSTRUCTION

Causal configuration classes

All quantities are unmeasured planning assumptions. Accelerator throughput, power limits and thermal configuration need qualified acceptance. Each rack observes space, peak power, weight and separation constraints; A/B feeds are not additive load.

Case / year	Site	CPU / GPU systems	Shelves / racks	Peak / typical kW	Durable TiB
base / 2027	RENO	3 / 11	5 / 11	96.4 / 60.05	150.0
base / 2027	BOISE	2 / 2	5 / 3	23.6 / 14.6	150.0
base / 2031	RENO	4 / 15	9 / 15	132.4 / 82.5	311.04
base / 2031	BOISE	2 / 3	9 / 4	34.8 / 21.6	311.04
base / 2036	RENO	5 / 22	21 / 22	198.8 / 123.95	773.97
base / 2036	BOISE	3 / 4	21 / 7	53.2 / 33.05	773.97
slower_adoption / 2027	RENO	3 / 7	5 / 7	64.4 / 40.05	90.0

Case / year	Site	CPU / GPU systems	Shelves / racks	Peak / typical kW	Durable TiB
slower_adoption / 2027	BOISE	2 / 2	5 / 3	23.6 / 14.6	90.0
slower_adoption / 2031	RENO	3 / 7	7 / 7	66.0 / 41.05	186.62
slower_adoption / 2031	BOISE	2 / 2	7 / 4	25.2 / 15.6	186.62
slower_adoption / 2036	RENO	3 / 6	13 / 6	62.8 / 39.05	464.38
slower_adoption / 2036	BOISE	2 / 2	13 / 4	30.0 / 18.6	464.38
high_demand / 2027	RENO	3 / 15	7 / 15	130.0 / 81.05	225.0
high_demand / 2027	BOISE	2 / 3	7 / 4	33.2 / 20.6	225.0
high_demand / 2031	RENO	4 / 36	13 / 36	303.6 / 189.5	466.56
high_demand / 2031	BOISE	3 / 5	13 / 6	54.8 / 34.05	466.56
high_demand / 2036	RENO	9 / 105	29 / 105	872.4 / 544.75	1160.95
high_demand / 2036	BOISE	4 / 12	29 / 12	124.4 / 77.5	1160.95
delayed_build / 2027	RENO	3 / 11	5 / 11	96.4 / 60.05	150.0
delayed_build / 2027	BOISE	2 / 2	5 / 3	23.6 / 14.6	150.0
delayed_build / 2031	RENO	4 / 15	9 / 15	132.4 / 82.5	311.04
delayed_build / 2031	BOISE	2 / 3	9 / 4	34.8 / 21.6	311.04
delayed_build / 2036	RENO	5 / 22	21 / 22	198.8 / 123.95	773.97
delayed_build / 2036	BOISE	3 / 4	21 / 7	53.2 / 33.05	773.97

Reference hardware and uncertainty

The sourced DGX H100 10.2 kW maximum is a separate sensitivity from the unqualified 8 kW capped design class. It is not an adopted supplier BOM or proof that the proposed throughput is attainable at a cap.

Site	Reference maximum kW	Racks	Initial envelope sufficient
RENO	120.6	11	False
BOISE	28.0	4	False

Throughput sensitivity	Site	GPU systems	Peak kW
low / 800 tokens/sec	RENO	20	168.4
low / 800 tokens/sec	BOISE	3	31.6
high / 2400 tokens/sec	RENO	8	72.4
high / 2400 tokens/sec	BOISE	2	23.6

Owned engineering arithmetic

These are component ratings for a concept, not a licensed protection, structural, fire or hydraulic design. Annual PUE does not replace coincident peak-load calculations.

Stage	Usable IT kW	Peak facility input kW	Surviving UPS path kW	N+1 generation / thermal cooling kW
initial	250	396.67	350	600 / 300

Stage	Usable IT kW	Peak facility input kW	Surviving UPS path kW	N+1 generation / thermal cooling kW
expanded	500	793.33	700	1200 / 600

Finance and workforce reconciliation

Phase I includes the $3M land overlay. No land payment is evidenced. The $15,500,000 envelope reconciles to $3,000,000 land, $780,000 unaccepted/unrecognized 2026 requests, $11,220,000 conditional post-2026 CIP requests and $500,000 unspent reserve. Earlier calibrated journals remain unchanged outside the separate land overlay. The enterprise successor applies existing finite Treasury limits to conditional requests.

Workforce phase	Technical FTE	Facilities / guards	Annual gross payroll	New authorized / occupied
COLO	20	0 / 0	$3,590,000	0 / 0
OWNED_CONDITIONAL	20	2 / 6	$4,530,000	0 / 0

Construction lead allocation is within the vendor-coordination pool; purchased specialist capacity is within design/commissioning costs. Neither is added again as permanent payroll. Workstations and roving assignments remain requirements, not occupancy records.

Base case year	Facility request	IT request	Operating request	Gross requirement
2026	3780000.00	0	0.00	3780000.00
2027	4795000.00	4360000	4401710.00	13556710.00
2028	6425000.00	720000	4672224.20	11817224.20
2029	0	280000	4822359.14	5102359.14
2030	0	464000	4983284.45	5447284.45
2031	0	6104000	5266189.81	11370189.81
2032	0	440000	5447171.81	5887171.81
2033	0	440000	5763525.54	6203525.54
2034	0	744000	5971606.16	6715606.16
2035	0	8168000	6323864.67	14491864.67
2036	0	1184000	6590391.31	7774391.31

Owned investment sensitivities separately include a demand-triggered second module, refurbishment, escalation, migration overlap and decommissioning. Cases above the 500 kW owned envelope remain infeasible; an attractive arithmetic NPV cannot approve an undersized facility. Terminal proceeds occur only in the terminal period. Tax benefits and verified displaced-cost credits are zero.

Recovery and execution gates

Native service	Tier	Minimum service
SVC-keys	BOOTSTRAP	Independent HSM pair, sealed recovery material and off-primary custodians; acknowledge key changes only after recovery durability
SVC-connectivity	BOOTSTRAP	Offline signed DNS/time/routing bootstrap; independent transport

Native service	Tier	Minimum service
		credentials
SVC-identity	BOOTSTRAP	Independent recovery realm and break-glass roots; no primary-only identity dependency
SVC-observability	BOOTSTRAP	Local protected incident log buffer before central collectors
SVC-compute	PRODUCTION	Prioritized transaction/database workloads
SVC-finance-systems	PRODUCTION	Minimum finance transaction/reporting dependencies; ICFR relevance requires customer scope
SVC-atlas-product	PRODUCTION	Prioritized centrally hosted client/professional planes; optional accelerators degraded
SVC-foundry-product	PRODUCTION	Central control/support only; customer operating authority retained
SVC-institutional	ALEXANDRIA	Canon authoritative records, minimum Pinakes/Semaphore discovery and source-rights enforcement; indexes rebuildable
SVC-private-ai	ALEXANDRIA	10% planning accelerator throughput; core institutional records do not wait for full AI restoration
SVC-developer	DEVELOPMENT	Rebuildable development; signed release artifacts protected with production prerequisites

Recovery is unverified for every tier. Prepositioned copies, independent keys/identities/DNS, complete logs and restore tombstones are required; a full network seed is not the timed minimum-service restore. The source register and accountable external evidence gates are in `enterprise/runtime/readiness.json`. Management owns implementation; Internal Audit retains independent evaluation.